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Quantum Secure Multiparty Computation based on Secure Summation and QKD

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Abstract

Secure Multiparty Computation (MPC) is a cryptography technique that allows multiple parties to securely perform operations on their private inputs without revealing them. It is used in various applications, such as data aggregation in the cloud, IoT, etc., where data privacy is the prime concern. Many researchers have explored ways to develop secure protocols with minimal risk of data leakage. Many of these classical secure MPC protocols depend on classical random number generators. However, advances in quantum information processing are putting classical key distribution methods at risk. Traditional summation protocols adopt an (n, n) threshold approach, requiring the participation of all n players to compute the sum securely. However, the proposed quantum secure multiparty computation based on secure summation and QKD (QSMPC-SSQKD) introduces a generalised quantum secure multiparty summation protocol. The proposed protocol utilises entanglement, a fundamental property of quantum mechanics, to facilitate safe key exchange, thereby ensuring robust security measures. We use a quantum random number generator (QRNG) to generate true random numbers, ensuring privacy preservation. The protocol achieves a per-bit entropy level of 0.708 through QRNG, which provides strong randomness. The security of the proposed protocol is validated using the CHSCH test ($\mathcal{B} \approx 2.8$), ensuring that any attempt to eavesdrop on the system can be detected, and the complete process will be rescheduled. Compared to some similar protocols, the proposed QSMPC-SSQKD protocol provides higher randomness and lower quantum circuit cost without using any trusted third party (TTP). This improvement enhances the protocol's efficiency and practicality in preventing data leakage during computation.

Keywords: Quantum secure multiparty computing, secure summation, quantum key distribution, Quantum random number, QPRNG, Bell's inequality, CHSH inequality, qiskit

1 Introduction

Secure multiparty computation is a critical concept that can be used by multiple parties to collaboratively compute a function over their respective parties secret data while preserving each other's private data from one another. Yao [1] first introduced MPC and is essential in real-time applications such as financial computation and data analysis in healthcare care where data aggregation is the primary concern. The objective of MPC is to provide a sure method to preserve the privacy of secret data from untrusted parties that collaborate on a computation.

Privacy is the primary concern for any party that shares their private data over the cloud to participate in data analysis applications like healthcare, finance, and the Internet of Things. In recent years, many solutions to these problems have been provided by researchers [1–5]. However, the classical protocols with broader applicability generally tend to be more complex or rely on the Computational Hardness Assumption for security.

One of the classical approaches to the MPC problem is the secure sum, which is used for data mining applications. Secure sum is a fundamental operation used in many privacy-preserving protocols by allowing untrusted parties to compute the sum of their private data jointly. The privacy-preserving secure sum ensures that no single party will get the original data of other parties, and yet participate in collaborative computing. To ensure the security of MPC, these protocols rely on classical cryptography methods like (Garbled Circuits [6], Oblivious Transfer [7], and Homomorphic Encryption [8]). These MPC protocols become vulnerable to quantum attack after Shor's [9] and Grover's [10] Algorithms.

In the context of secure summation protocols for privacy preservation, quantum techniques can offer significant security benefits by introducing Quantum Key Distribution (QKD) as shown in the study [11, 12] work. The first QKD protocol was introduced by Bennett and Brassard's [13], which has a revolutionary impact on cryptographic protocols. After BB84 [13], Artur K. Ekert [14] has presented novel methods for key exchange that rely on the entanglement principle, which is based on Bell's theorem, [15] for key security. To secure the privacy of MPC protocol form the external adversary E91 [14] QKD protocol protocol can provide unconditional.

The proposed QSMPC-SSQKD protocol provides a quantum secure privacy-preserving model for the summation protocol. The proposed model uses the E91 protocol along with a quantum random number generator (QRNG). Initially, all the parties perform key distribution using the E91 QKD protocol to ensure security against adversaries outside collaborative computing. The QRNG uses the quantum mechanics properties to provide truly random numbers to preserve the privacy of the protocol initiating party $Alice_1$. After collaborative computing, the final secret is shared with all parties by encrypting with the secret key k_i of each party. The security of the key

is based on Bell's inequality, so the parties with corresponding key k_i will come to know if any Eve tries to eavesdrop on the key.

The main contributions of the proposed protocol are as follows:

- The protocol provides a quantum secure privacy-preserving method for secret sharing in QSMPC-SSQKD using a secure sum method.
- For secure sum, the protocol uses true random numbers generated using the components of quantum mechanics.
- The protocol provides a security bound to preserve the privacy of the semi-honest parties collaborating in secure computing.
- Security analysis of the proposed protocol in terms of an external adversary monitoring the key exchange process.

The organisation of the paper is as follows: the section 2 provides similar work on secure MPC protocols and the QSMPC-SSQKD protocol in recent years. Section 3 gives a brief overview of the E91 protocol and QRNG with a simple Qiskit circuit. Section 4 proposes the QSMPC-SSQKD protocol using a secure sum approach. The security analysis of the proposed work is described in the section 5. Section 6 provide the result and comparison of the proposed work. Section 7 provides an application scenario along with its theoretical and practical principles of the protocol. Section 8 summarises the proposed work, its limitations, and future scope.

2 Related work

Li et al. [16] have proposed a secure summation protocol for multiparty computation using quantum mechanics properties. They used inverse Quantum Fourier Transform (QFT), which is a phase operator used to rank the data securely. The circuit they used does not specify the encoding of private input data. After encoding, they have created a superposition state of the qubit, which reveals no information. The proposed system complexity is too complex, as the inverse QFT requires a higher implementation cost.

Clifton et al. [17] have proposed a model to deal with privacy-preserving during data mining. They have presented four different methods for MPC, including the secure sum method. They have used the secure sum method, which is frequently used in distributed data mining to calculate the sum of values from the individual parties. They have assumed that no party colludes to get others' private data. They have also utilised a classical random number to calculate the sum and follow a closed path, which is permuted for every run to change the neighbours of the parties. Still, any external adversary monitors the part each time and can get the details of each neighbour and their data. It needs to be secure from external adversaries.

Jande et al. [2] have proposed a hybrid version of the secure sum [17] protocol. They have used a third trusted party (TTP), which takes secret data of all parties, including their random number. TTP performs a sum over the data and sends the sum to all the parties. All parties remove their first random number, add another part of the data with another random number, and send it back to TTP. TTP and all the parties repeat the steps until all the data is added. Finally, TTP will send the sum of all the data to all the parties after removing all random numbers. The basic

assumption is that TTP is a trusted party, which is doubtful without considering the ideal environment. If a trusted party exists, then there is no requirement for MPC in real-time. Hence, the protocol has a single point of failure: TTP. Bhanumathi et al. [18] uses the same concept as [2] for the experiment with three parties.

Ding et al. [3] have proposed a secure MPC protocol using a distributed Kalman filtering scheme. The Kalman filtering scheme provides security of the private data from the adversary without altering the performance of the filtering process. They have used secure summation [2] integrated with the Kalman filtering process. Instead of a closed network (ring network), they have used a group-based design network to ensure privacy against data theft. They have used public key cryptography to secure the data distribution over the network, which is vulnerable to quantum attacks. The security of classical public key cryptosystems (like RSA, ECC) depends on factorisation or discrete logarithms, proven to be insecure after Shor's algorithm [9, 19].

To secure MPC from quantum attack, Li et al. [20] have proposed a QMPC protocol that is capable of identifying cheaters within the communicating parties. They have used concepts of quantum machines such as the d-dimensional GHZ (Greenberger-Horne-Zeilinger) states protocol [21] along with Shamir's threshold scheme [20] and hash function to identify the cheater in the QMPC. They have used a summation approach to compute the secret value without revealing the secret data of individual parties. Using quantum machines, QMPC provides security against intercept and resend (IR) attacks, entanglement measure attacks, and malicious attacks by semi-honest TTP. One major issue in this protocol is that if any party provides incorrect data, then all parties will receive incorrect results. To overcome this limitation, they have proposed that each party's secret length be > 4. Still, the protocol is vulnerable to a quantum attack if an external adversary is monitoring the system.

Sutradhar et al. [22] have proposed a QMPC protocol using secure summation to overcome the limitation of the classical summation MPC protocol. They have determined the (k, n) threshold for secure communication, where k out of n parties can securely perform computation without leaking the privacy of other parties. They have used a random vector along with their secret data to perform a secure summation. Each party in the protocol uses an entangled state and applies the QFT [23] to the first qubit. All the parties share the encoded secret data in these entangled states for summation purposes. The protocol provides security against IR attacks, entangled measure attacks and collusion attacks. Even if dishonest parties collude, they will not retrieve the secret data of other parties. The protocol is limited; if more than $(n - k)$ parties collude together, the secret of other parties can be compromised. Additionally, the protocol's operation cost will be high due to entanglement and QFT operations.

Liu et al. [24] have proposed a secure quantum scalar product (S2QSP) protocol for secure MPC. The protocol computes the scalar product between two parties by utilising properties of entangled states and QFT. The scalar product uses a sum to implement random numbers in bits, which provides security against measurement attacks. And QFT provides polynomial complexity in S2QSP. The protocol provides an entanglement bondage method to verify the honesty of parties, which ensures security against forgery and dishonest behaviour. The implementation cost of entanglement

bonds and sums may be very high. They have used an IBM Qiskit simulator for the feasibility test, which may vary with implementation on actual quantum hardware.

3 Preliminaries

3.1 E91 Protocol

E91 protocol [14] was first introduced by Arthur Ekert in 1991 using the quantum machine property called quantum entanglement. E91 protocol is used to share secret keys between the two parties using two spin-$\frac{1}{2}$ singlet states. The protocol ensures key safety by detecting any attempts to disrupt the key distribution process. The Fig. 1 shows E91 protocol structure where two parties $Alice_1$ and $Alice_2$ taking part in secure key distribution. The source generates a two-spin-$\frac{1}{2}$ singlet state and shares one singlet state to $alice_1$ and another to $Alice_2$ where both parties have two choices of measurement basis: X_b and Z_b. After receiving the entangled particles, both the E91

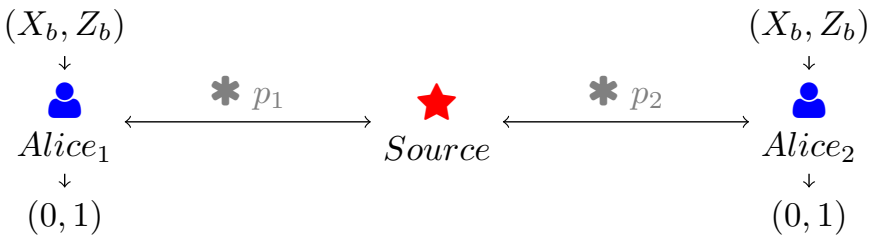


Fig. 1 QKD using E91 protocol between $Alice_1$ and $Alice_2$

protocol, which is as follows:

Step 1 : As shown in the Fig. 1 the source shares two entangled particles (p_1 and p_2) of state $|\psi\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$ among $Alice_1$ and $Alice_2$.
Step 2 : Upon receiving respective particles, $Alice_1$ and $Alice_2$ randomly select measurement setting from the observables $X_b = \sigma_x$ or $Z_b = \sigma_z$ for the measurement. After measuring the particles, each party stores the result as the binary outcome (0, 1). Once the measurement process is over, each party shares its choice of measurement basis with each other using the classical channel.
Step 3 : The parties $Alcie_1$ and $Alice_2$ know about the measurement choice of each other for every respective particle sequence they received. Both parties created two groups based on the measurement results they obtained. The particles for which the measurement basis is different were stored in the decoy state group (D_g, and the particles for which the measurement basis is the same are stored in the key state group (K_g).
Step 4 : The group D_g is used to detect the presence of the adversary (Eve) in the system. In an ideal environment, any error caused by Eve makes him entangled with the particles of the group (D_g). If there is any error in D_g, it ensures the presence of Eve in the system. Hence, both parties conclude that eavesdropping has occurred in the system, and they restart the key distribution process immediately.

Step 5 : If the parties find the same quantum channel, they use K_g for the key. Both $Alice_1$ and $Alice_2$ used the same measurement basis for the particles in the group K_g, and K_g will be used as a key. As shown in the Fig. 1 after the measurement, the output state $|0\rangle$ and $|1\rangle$ are represented as classical bit 0 and 1 respectively.
Step 6 : The protocol ends successfully.

3.2 Quantum random number

QRNG uses quantum mechanics property to generate a truly quantum random number Q_r, unlike deterministic classical random number generators. QRNG uses the concept of superposition. A superposition state has an equal probability of collapsing into either of two possible states(let's say state $|0\rangle$ or $|0\rangle$). A lot of work has been done in the field of QRNG, which can be studied in the work of Collantes et al. [25].

QRNG's randomness results from the fact that the measurement of superposition is not deterministic. For instance, let's consider a quantum particle is prepared in a superposition state ($\frac{|0\rangle+|1\rangle}{2\sqrt{2}}$) using hedamard gate (H), a measurement of its polarization will yield a random outcome, either $|0\rangle$ or $|1\rangle$, with equal probability. A simple 8 bit QRNG is shown in the Fig. 2. The circuit is created using the IBM Qiskit simulator (Qiskit-1.2.1; Python 3.10.12; Google Colab), which is used to run quantum algorithms on quantum computers and cloud simulators of Google.

The QRNG shown in Fig. 2 uses 8 qubits and all qubits are operated under H to prepare the superposition state of all the qubits. After the measurement, each qubit will be collapsed into random states, and their measurement outcome will be stored in a classical register of 8 bits. Whenever the circuit complies, it generates a truly random number, which provides unconditional security against any attack.

Min-entropy is used to quantify the randomness of the QRNG in an untrusted environment. Let the a QRNG process is (Pro), in the worst case min-entropy (H_{Pro}) gives the uncertainty about the outcome (y) of the process which is given by:

$$H_{Pro} = -\log_2 \left(\max_x p(y) \right) \tag{1}$$

Under the ideal system scenario, the measurement will give an outcome (y) with equal probability; hence, H_{Pro} is maximised, implying perfect randomness.

4 Proposed work

The proposed work provides a secure protocol for secret sharing among n parties. The proposed work includes key distribution followed by secret sharing in a cyclic manner. The objective of the proposed protocol is that no party will know the other's secret data except the final data.

4.1 Key Distribution

The proposed protocol include n parties ($Alice_1$, $Alice_2$, $\cdots$, $Alice_i$, $\cdots$, $Alice_n$). The Fig. 3 shows the key sharing using the E91 protocol (provided in Section 3.1). Here, each party establishes a secure key using the E91 protocol. As shown in Fig. 3 $Alice_1$

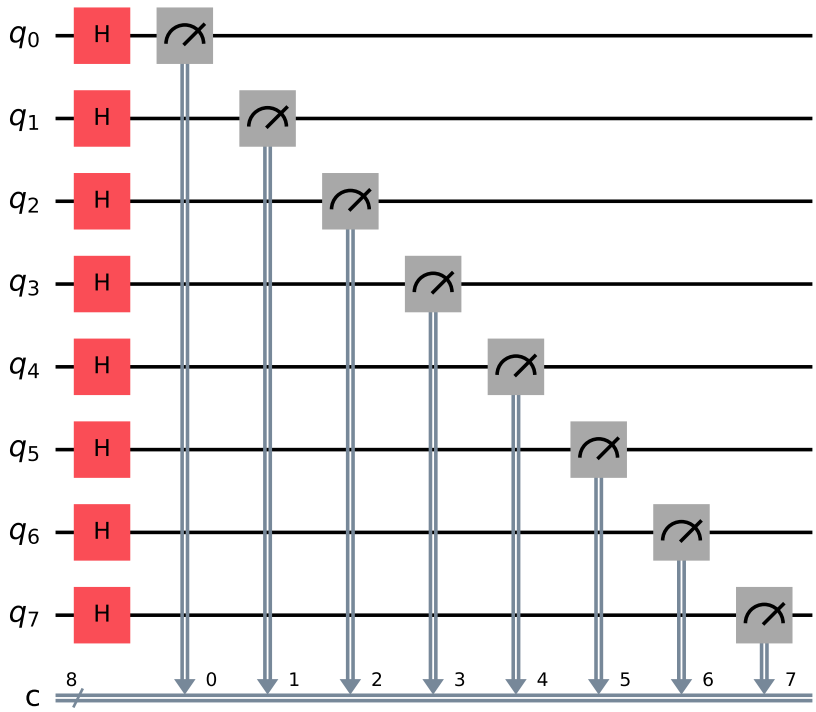


Fig. 2 8 bit QRNG circuit generated using IBM Qiskit, where H is Hadamard gate, $q_0, q_1, \cdots, q_7$ are quantum registers which store qubits, C is 8-bit classical register to store the measurement outcomes

and $Alice_2$ share secret key $k_{1,2}$, $Alice_1$ and $Alice_3$ share the key $k_{1,3}$, similarly all the parties $Alice_i$ and $Alice_n$ share the key $k_{i,n}$, where $i \in n$.

Using E91 protocol provides unconditional security against eavesdropping. Under the ideal condition where no external noise is present in the system, the error is caused by the presence of Eve in the system. Hence, there will be a bit error during the key exchange, which reveals the presence of Eve to the $Alice_i$ and $Alice_k$, where $k \in n$. And the key distribution will be rescheduled between the parties $Alcie_i$ and $Alcie_k$. In an ideal scenario, when Eve tries to intercept the key exchange process, some amount of information can be revealed to him. According to Li et al. [26], Eve can get maximum information during eavesdropping, which is given as:

$$I(A_i, E_v) = H(A_i) - H(A_i|E_v) \tag{2}$$

As the particle are randomly shared by $Alice_i$, hence $H(A_i) = 1$ and

$$H(A_i|E_v) = -\sum p(A_i, E_v) \log_2 p(A_i|E_v) \tag{3}$$

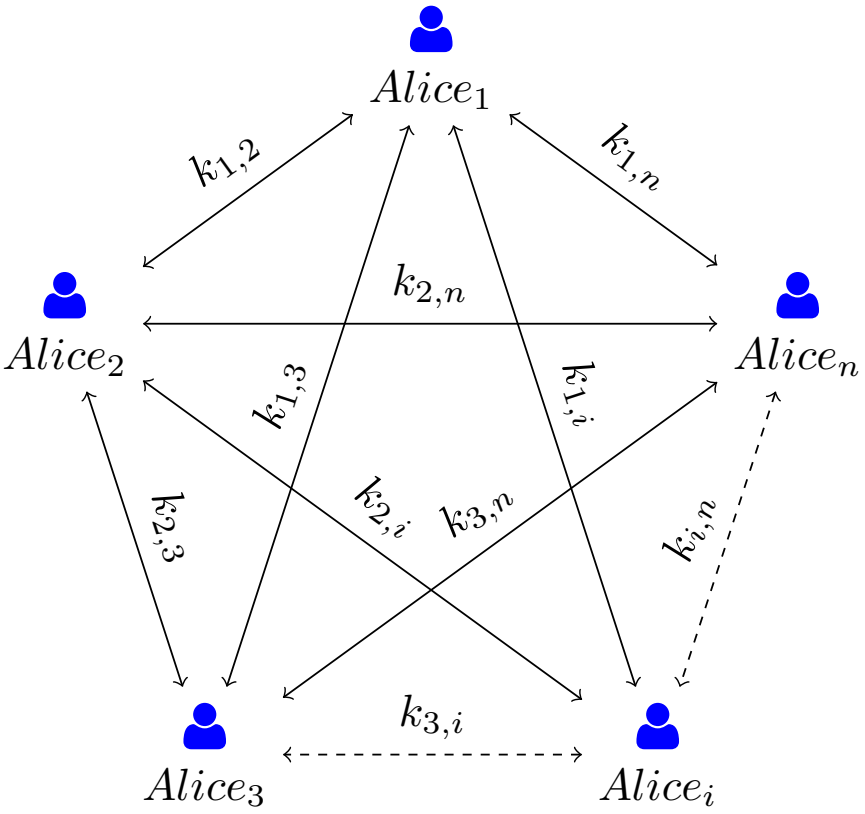


Fig. 3 Secure key Sharing between each $Alice_i$ using the E91 QKD protocol (Entanglement based), where $k_{i,n}$ is the secret key created between the $Alice_i$ and $Alice_n$, and $(i, n) \in \{0, 1, 2, 3 \cdots, n\}$

From Eq. (2), $Alcie_i$ can be related to the maximum value of noise during the key exchange process under an ideal environment scenario.

4.2 Proposed secret sharing model

In the proposed model, each party has secret data m_i, and all parties need the combined data of the other parties without revealing their own secret data. Fig. 4 shows a secret sharing mechanism to share secret data using a quantum secret key.

The procedure of secret sharing is as follows:

Step 1 : The one-to-one key distribution is initiated among all parties using the E91 protocol as shown in Fig. 3 in section 4.1.
Step 2 : Generation of Q_r only by the initiating party of the protocol $Alice_1$ as shown in section 3.2.
Step 3 : $Alice_1$ compute:

$$M_1 = m_1 + Q_r \tag{4}$$

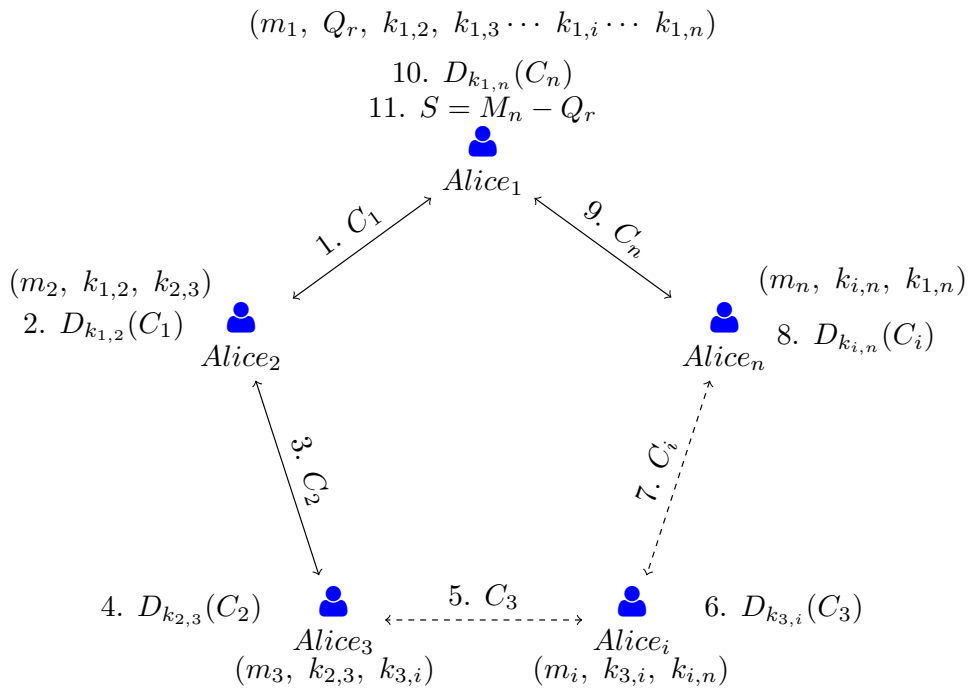


Fig. 4 Secure communication of secret data in a ring structure. Here, $Alcie_1$ initiate the protocol suing the QRNG (Q_r) and its secret message m_1 and the respective key of adjacent negihbours, here $C_1 = E_{k_{1,2}}(M_1)$; $C_2 = E_{k_{2,3}}(M_2)$; $C_3 = E_{k_{3,i}}(M_3)$; $C_4 = E_{k_{i,n}}(M_i)$ and $C_5 = E_{k_{1,n}}(M_n)$.

Step 4 : $Alice_1$ forward $C_1 = E_{k_{1,2}}(M_1)$ to next party $Alice_2$ using there secret key $k_{1,2}$. $Alice_2$ decrypt the received massage $D_{k_{1,2}}(C_1)$ using the same key and compute:

$$M_2 = M_1 + m_2 \tag{5}$$

Step 5 : $Alice_2$ encrypt M_2 using the secret key $k_{2,3}$ and send $C_2 = E_{k_{2,3}}(M_2)$ to $Alice_3$.
Step 6 : Similarly, $Alice_i$ decrypt the message $D_{k_{3,1}}(C_3)$ and compute:

$$\begin{aligned} M_i &= M_{i-1} + m_i \\ M_i &= Q_r + m_1 + m_2 + m_3 + \cdots + m_i \end{aligned} \tag{6}$$

where $i \in \{1, 2, 3, \cdots, n\}$, and send $C_i = E_{k_{i,n}}(M_i)$ to $Alice_n$.
Step 7 : Finally $Alice_n$ decrypt $D_{k_{i,n}}(C_i)$ send $C_n = E_{k_{1,n}}(M_n)$ to $Alice_1$.

After completion of the secret data sharing process, the initiating party of the protocol $Alice_1$ performs the final secret (S) sharing, which is shown in Fig. 5. The process of sharing S is as follows:

Step A : $Alice_1$ decrypt $D_{k_{1,n}}(C_n)$ using the secret key $k_{1,n}$ and obtain M_n.

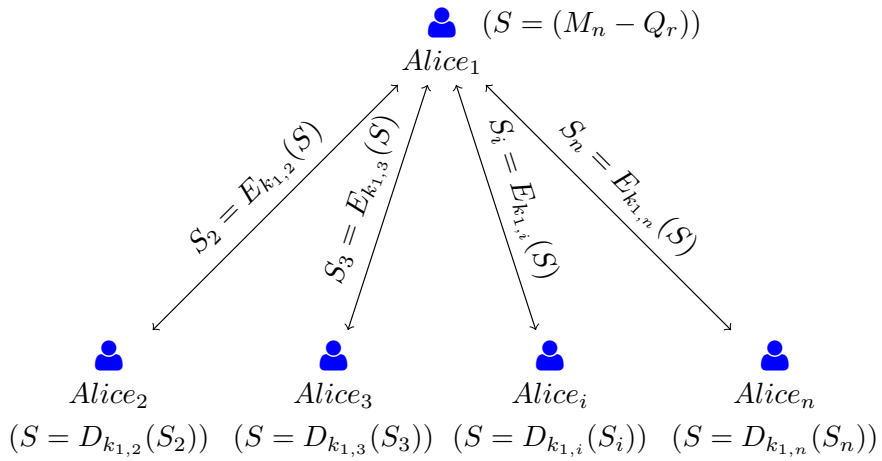


Fig. 5 secure sharing of final data

Step B : $Alice_1$ compute S

$$S = M_n - Q_r \tag{7}$$

Step C : Finally, S will be distributed to each party by encrypting with their respective secret keys.

Here, S is the final data that all parties need for their analysis while ensuring the privacy of each other's private data. Hence, the proposed model provides unconditional security of individual private data from external Eve. The protocol guarantees the privacy-preserving of individual private data data in the presence of Eve.

5 Security Analysis

The security of the proposed system relies on two cases, which are as follows:

Case 1 : External factors (adversary) trying to reveal the secure key used in the protocol as given in 5.1
Case 2 : When internal parties (semi-honest) get corrupted as shown in 5.2.

5.1 Security against quantum adversary

Case 1: The security of the key is based on the strength of the E91 QKD protocol. Section 3.1 provides the working of the E91 protocol. Its security is derived from the violation of Bell's inequality, which follows the basic assumption that $Alice_i$ and $Alice_{i+1}$ share a maximally entangled bell state $|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_{A_i}|0\rangle_{A_{i+1}} - |1\rangle_{A_i}|1\rangle_{A_{i+1}}\right)$ where $|0\rangle$ and $|1\rangle$ represent two orthogonal quantum states, and A_i and A_{i+1} refer to $Alice_i$ and $Alice_{i+1}$, respectively and where $i \in \{1, n\}$.

Let us demonstrate the security based on the Clauser-Horne-Shimony-Holt (CHSH), an extension of Bell's inequality. The two choices of measurement basis (X_b, Z_b) for the two parties are shown in Fig. 1. $Alice_i$ and $Alice_{i+1}$ the following measurement setting:

- $Alice_i$ uses $A_i^1 = X_b$ and $A_i^2 = Z_b$
- $Alice_{i+1}$ uses $A_{i+1}^1 = \frac{X_b + Z_b}{\sqrt{2}}$ and $A_2^2 = \frac{X_b - Z_b}{\sqrt{2}}$

Now, let's consider Bell-CHSH inequality:

$$\begin{aligned}\mathcal{B} = |E(A_i^1, A_{i+1}^1) + E(A_i^1, A_{i+1}^2) + E(A_i^2, A_{i+1}^2) \\ -E(A_i^2, A_{i+1}^2)|\end{aligned} \tag{8}$$

where $E(A_i^x, A_{i+1}^x)$ shows correlation among the observables of $Alcie_i$ and $Alcie_{i+1}$. Following the Eq. (8) offer the maximally entangled state $|\psi\rangle$ the quantum value $\mathcal{B}$ will be $2\sqrt{2}$. The value $2\sqrt{2}$ guarantees optimal security that any attempt to intrude on the key distribution process by Eve is detected by $Alice_i$ and $Alice_{i+1}$, and the key distribution will be rescheduled. If the $\mathbf{B} \neq 2\sqrt{2}$ it colludes with the following security impacts

- Eve has tried to introduce the entanglement by destroying it or by taking measurements of the hidden state, due to which the system is vulnerable to possible attacks such as intercept and resend attacks.
- The key rate will be reduced, and Eve will get some information from Eq. (2).

Fig. 6 shows the circuit diagram of CHSH performed using qiskit. And Algorithm 5.1 provides the pseudo code of the CHSH inequality test using the Bell state. Here, $Alice_i$ perform two Hadamard operations and $Alice_{i+1}$ perform rotation of the qubit about the y-axis using the $R_y(\pi/2)$ gate followed by the $R_z(\pi/2)$ gate. This guarantees that the measurement on the computational basis is comparable to projecting on a custom basis that is relevant to the CHSH test (for example, at angle $3\pi/8$).

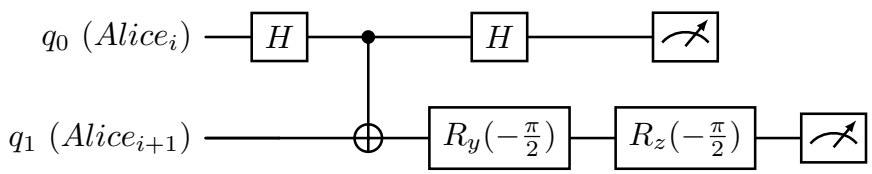


Fig. 6 Circuit diagram of CHSH

Algorithm 1 CHSH Value using Bell State Simulation

function COMPUTECHSH(N)
  Initialize $results \leftarrow \{\}$, $backend \leftarrow$ qasm_simulator
  for $a \in \{A_i^1, A_i^2\}$ **do**
    for $b \in \{A_{i+1}^1, A_{i+1}^2\}$ **do**
      Create 2-qubit circuit qc; apply $H(0)$, $CX(0,1)$
      if $a = A_i^2$ **then** apply $H(0)$
      end if
      if $b = A_{i+1}^1$ **then** apply $R_y(-\pi/2), R_z(-\pi/2)$ on qubit 1
      else if $b = A_{i+1}^2$ **then** apply $R_y(\pi/2), R_z(\pi/2)$ on qubit 1
      end if
      Measure all qubits; run on $backend$ with N shots
      Store result counts in $results[a_b]$
    end for
  end for
  $E \leftarrow \{\}$ ▷ Expectation values
  for all $(k, c) \in results$ **do**
    $E[k] \leftarrow \frac{1}{\sum c} \sum_{x \in c} c[x] \cdot (1 \text{ if } x \in \{00, 11\} \text{ else } -1)$
  end for
  $\mathcal{B} \leftarrow E(A_i^1, A_{i+1}^1) + E(A_i^1, A_{i+1}^2) + E(A_i^2, A_{i+1}^2) - E(A_i^2, A_{i+1}^2)$
  return $(\mathcal{B}, E)$
end function

From each of the expectation values ($E(A_i^1, A_{i+1}^1), E(A_i^1, A_{i+1}^2), E(A_i^2, A_{i+1}^2)$, and $E(A_i^2, A_{i+1}^2)$) the Bell-CHSH value $\mathbf{B} \approx 2.8$ is greater than 2. This violation results in a clear demonstration of entanglement between two parties.

Hence, the protocol will be secure from any external attempt to intrude on the key. Preserves the privacy of individual private data in MPC.

5.2 Security against semi honest parties

Case 2: Consider an MPC with four parties $Alice_1$, $Alice_2$, $Alice_3$, and $Alice_4$. Now, there may be three possible scenarios in which an internal MPC party gets corrupted.

Scenario 1: Let the protocol initiating party $Alice_1$ be dishonest and try to obtain the other parties' private data. After completion of data sharing, $Alcie_1$ will have $S = m1 + m_2 + m_3 + m_4 + Q_r$. After removing his data (m_1, Q_r) from S $Alice_1$ will gain the information IG:

$$\begin{aligned} IG &= S - (m_1 + Q_r) \\ &= m_2 + m_3 + m_4 \end{aligned} \tag{9}$$

From Eq. (9), it is clear that if one party, either the protocol initiating party, becomes dishonest, it will know nothing about the private data of the other parties.

Scenario 2: In the second scenario, let two parties $Alice_1$ (the protocol initiating party) and $Alice_3$ collude to get the private data m_2 and m_4 of $Alice_2$ and $Alice_3$ respectively. Let $Alice_3$ receive a message ($M_2 = Q_r + m_1 + m_{@}$) forwarded from

$Alice_2$. As $Alice_3$ is colluded with $Alice_1$ they have the information of their data (m_1, m_2, Q_r) and perform computation to get data of $Alice_2$:

$$\begin{aligned} IG_{A_2} &= M_2 - (m_1 + m_3 + Q_r) \\ &= (m_1 + m_2 + m_3 + Q_r) - (m_1 + m_3 + Q_r) \\ &= m_2 \end{aligned} \tag{10}$$

Similarly, after the final process as shown in Eq. (10), both parties will get $Alice_4$'s private data m_4. *Scenario 3*: If the internal party willingly provide wrong data during the process, then the final S shared to all the parties has useless information.

As a result, the QSMPC-SSQKD protocol is sure to be under the first scenario, where only one internal party is allowed to be corrupt. For n parties in the system, if $\left(\left\lceil \frac{n-4}{2} \right\rceil + 2\right)$ parties collude together, the complete data of all parties will be disclosed, where $n \geq 4$.

6 Result and Comparison

This section demonstrates moderate QRNG entropy strength, secure entanglement through the CHSH test, and low-latency performance in simulation of the proposed QSMPC-SSQKD protocol. It ensures unpredictability, resistance to eavesdropping, and strong input confidentiality. Compared to existing schemes, it achieves quantum-level security with reduced circuit complexity and better feasibility.

6.1 QRNG exterminate result

After running the QRNG circuit over 1024 shots the most frequent output is achieved 20 times. Hence, the probability $max(p_y) = \frac{20}{1024} = 0.01953$. Applying it to the Eq. (1) we will get the H_{pro}:

$$H_{\text{Pro}} = -\log_2(0.01953) \approx 5.67 \text{ bits} \tag{11}$$

To classify the strength of the randomness, convert min-entropy to per-bit entropy Eq. (12):

$$H_{\text{per-bit}} = \frac{H_{\text{Pro}}}{n} = \frac{5.67}{8} \approx .708 \tag{12}$$

Table 1 provides strength categories of randomness along with suitable application.

Table 1 Min-Entropy Strength Categories

$H_{\textbf{per-bit}}$	**Strength**	**Applications**
≥ 0.98	Strong	Cryptographic applications
$0.7 - 0.98$	Moderate	Simulations, randomized algorithms
< 0.7	Weak	Needs extractor (e.g., classical: SHA-256)

Although the randomness falls under moderate strength, but using it with the E91 protocol guarantees strong security of the mask input of individual party.

The proposed QSMPC-SSQKD uses QRNG to provide unpredictability of Q_r, due to which reverse engineering of the input is not possible. Hence, it provides protection even under full network surveillance. Even if it offers true randomness, which ensures strong hiding capability of private input. Table 2 provide the security benefits of using QRNG in QSMPC-SSQKD.

Table 2 Security benefits of QRNG

Security benefits	**QRNG Contribution**
Input confidentiality	Unpredictable masking
Resistance to reverse engineering	QRNG output shows no pattern.
Eavesdropping protection	Nothing is leaked even with full traffic
High entropy masking	Ensures strong hiding guarantees

6.2 CHSH test result

After running the Algorithm 5.1 using qiskit the experiment value obtained is $\mathcal{B} \approx 2.8 > 2$. This confirms that the quantum nature of the entangled state for any $\mathcal{B} < 2.5$ would indicate possible eavesdropping.

6.3 Entropy-Based Leakage Estimation

Consider the Eq. (2), where $H(A_i)$ be $Alice_i$'s entropy and $H(A_i|E_v)$ be the entropy conditioned on Eve's knowledge. Now, assuming binary entropy with noise ϵ, the $H(A_i|E_v)$ will be:

$$H(A_i|E_v) = -\epsilon \log_2 \epsilon - (1-\epsilon) \log_2 (1-\epsilon) \tag{13}$$

When $\epsilon = 0.1$, $I(A_i : E_v) \approx 0.53$ bits. Hence, Eve's presence is more noticeable through CHSH deviation with a higher error rate.

6.4 Performance analysis result

The QSMPC-SSQKD protocol performance is based on the computational and practical feasibility of the protocol. Taking into account n the number of participants in the protocol, the establishment of keys between each participant is controlled by the E91 protocol, resulting in a total cost of $\binom{n}{2} = O(n^2)$ during key establishment. Each participant encrypts and decrypts their partial sum using their secret key, incurring a complexity of $O(n)$ cost for each process.

QRNG reliably produces an 8-bit string of random numbers, as illustrated in Fig. 2. The piratical demonstration of 8-qubit circuits with 1024 shots completes in under 2 seconds, providing better feasibility of the protocol in simulated environments.

Furthermore, the CHSH inequality is violated with a measured value of $S \approx 2.8$, confirming the presence of a secure quantum entanglement. For real-world deployment, it requires quantum channels and entangled photon sources, which are attainable through existing quantum cloud platforms.

6.5 Comparison

The comparative analysis of the QSMPC-SSQKD protocol with three related QMPC schemes: those by Jangde et al. [2], Sutradhar et al. [22], and Li et al. [16] is provided in Table 3. Regarding key establishment, Ref. [2] relies on classical mechanisms, which are inherently vulnerable to quantum attacks. Ref. [22] uses GHZ-state entanglement and QFT, which significantly increase circuit complexity. Li et al. [16] employ a combination of entanglement and cryptographic hash functions, offering quantum-level protection at a high computational cost. In contrast, the proposed QSMPC-SSQKD protocol adopts the E91 protocol based on Bell inequality violations, achieving quantum-secure key establishment with relatively lightweight requirements.

In order to provide randomness, Ref. [2] uses pseudo-random number generators (PRNGs), which are deterministic and less secure. Similarly, Ref. [22] rely on classical randomness, while Li et al. [16] incorporate quantum randomness into their design. QSMPC-SSQKD protocol stands out by using a true QRNG, ensuring highly unpredictable values.

Collusion resistance is another critical criterion. Ref. [2] offers no formal threshold to guard against colluding parties. Sutradhar Ref. [22] supports a well-defined (k, n) threshold, allowing the system to tolerate up to k colluding parties. Li et al. [16] provide limited cheater detection but without a formal threshold guarantee. The proposed QSMPC-SSQKD protocol improves on this by defining a formal collusion threshold, ensuring provable confidentiality by $\left(\left\lceil \frac{n-4}{2} \right\rceil + 2\right)$ even under insider threat scenarios.

When it comes to quantum circuit complexity, Ref. [2] achieves low cost due to its classical approach. Whereas, Ref. [22] and Ref. [16] both have higher circuit complexity due to entanglement and QFT. In contrast, the proposed QSMPC-SSQKD protocol maintains low quantum circuit cost by leveraging Bell pairs for key establishment.

Finally, in terms of trust assumptions, Ref. [2] require a TTP to manage certain aspects of the protocol, which introduces a centralised point of trust and single point of failure. The other three protocols, including QSMPC-SSQKD, eliminate the need for a TTP, enhancing decentralisation and trust minimisation. Overall, the proposed QSMPC-SSQKD demonstrates a well-balanced design that combines quantum security, formal theoretical guarantees, low resource consumption, and decentralisation.

7 Discussion

QSMPC-SSQKD protocol uses the quantum mechanics property for secure computing, which has a significant impact on the application, theoretical, and practical domains. The protocol allows multiple parties to compute a summation on their private input without revealing it. The protocol is useful to provide secure computing in applications such as privacy-preserving data aggregation in healthcare to compute the total number

Table 3 Comparison with Related Quantum MPC Protocols (Transposed)

Feature	**Jangde et al. [2]**	**Sutradhar et al. [22]**	**Li et al. [16]**	**QSMPC-SSQKD (Ours)**
Key Exchange	Classical	GHZ/QFT	Entanglement + Hash	E91 (Bell-based)
Randomness Source	PRNG	Classical	N/A	QRNG
Threshold	None	(k, n) Threshold	Limited Cheater Identification	derived in section 5.2
Quantum Circuit Cost	Low	High	High	Low
TTP required	Yes	No	No	No

of patients with a specific condition without sharing individual patient data. Similarly, in the banking system the proposed protocol can be useful in computing aggregate transaction statistics without sharing individual transaction data. In addition, it can be useful in quantum networks and quantum cloud computing, where data aggregation is required while maintaining the privacy of data.

The protocol has important theoretical implications in addition to its real-world uses in several fields. It uses quantum resources, namely maximally entanglement qubits, to present a new method for safe computation. The protocol leverages quantum mechanics to create a secure key between two parties that is resistant to outside threats. To improve computational security, it also integrates a quantum random number generator (QRNG) based on inherent quantum features. This approach opens the door for more research on quantum protocols that may be used for a variety of computations other than summation.

In addition, the protocol demonstrates significant practical applicability by enabling integration with classical computing systems while relying on the inherent security properties of quantum mechanics. The protocol avoids the need to encode secrets on qubits, hence it reduces the error correction during private data sharing, which will make it practically feasible for the current quantum hardware, along with classical computers. The protocol scales linearly with the number of parties n, as each party performs QKD and secure summation. This makes it suitable for applications involving a large number of participants, such as in distributed sensor networks or Internet of Things [27, 28] systems. Overall, QSMPC-SSQKD offers a decentralised, quantum-secure, and computationally efficient framework for multiparty computation, paving the way for secure and scalable applications in quantum networks.

8 Conclusion

The proposed QSMPC-SSQKD protocol effectively preserves the privacy of secret data contributed by participating parties in a collaborative computation. By leveraging a secure sum approach masked with Q_r, the final computed secret S is shared securely among parties using individual secret keys established via the E91 QKD protocol. The protocol achieves a moderate per-bit entropy level of 0.708 through QRNG, ensuring sufficiently strong and unpredictable randomness suitable for secure applications.

Security is further reinforced through the use of Bell's inequality during the key establishment phase. This is validated by the result of the CHSH test ($\mathcal{B} \approx 2.8$), which confirms the presence of entangled quantum states and is used to detect the presence of any adversary.

Furthermore, entropy-based leakage analysis demonstrates limited information gain for an adversary under noisy conditions. The protocol supports a formal collusion threshold, remaining secure even when up to $\left(\left\lceil \frac{n-4}{2} \right\rceil + 2\right)$ semi-honest parties collude, ensuring strong confidentiality guarantees. While the protocol provides robust protection against passive threats, handling malicious behaviour, such as the injection of incorrect information by semi-honest parties, requires further extensions.

Declarations

Ethical Approval

No human and/ or animal studies have been presented in the manuscript. Hence, no ethical approval is needed.

Competing interests

The authors do not have any financial or personal conflicts of interest to disclose related to this manuscript.

Authors' contributions

Mandeep Kumar has drafted the manuscript and generated all tables. Bhaskar Mondal has generated all the figures. Both authors have worked on literature surveys, simulations, and proofreading.

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Availability of data and materials

No data was generated during the research to disclose. No Code is available to share.

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